

FINAL-YEAR B.TECH PROJECT REPORT

VDCET Official Website with AI-Powered RAG Knowledge Assistant

****STUDENT & TEAM DETAILS****

- **Project Lead:** Suraj Gajanan Pathade

- **Team Members:**

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- **Department:** Department of Computer Engineering

- **Degree:** Bachelor of Technology (B.Tech)

- **College:** Vilasrao Deshmukh College of Engineering & Technology, Mouda (VDCET)

- **Affiliated University:** Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere

- **Project Guide:** Rohini Ma'am

- **Academic Year:** 2026–2027

1. COVER PAGE & TITLE

PROJECT TITLE: VDCET Official Website with AI-Powered RAG Knowledge Assistant

INSTITUTE: Vilasrao Deshmukh College of Engineering & Technology (VDCET), Mouda, Nagpur (DTE Code: 04141)

UNIVERSITY: Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere, Maharashtra

2. CERTIFICATE OF APPROVAL

This is to certify that the project entitled “**VDCET Official Website with AI-Powered RAG Knowledge Assistant**” submitted by **Suraj Gajanan Pathade, Rohit Shirbhate, Khushal Hajare, Harshal Kalbhut, Sanket Nasare, and Tejas Sawargaokar** to the Department of Computer Engineering, Vilasrao Deshmukh College of Engineering & Technology, Mouda (VDCET), is a bonafide record of work carried out under the guidance of **Rohini Ma’am** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Computer Engineering** affiliated with **Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere** during the academic year 2026–2027.

Project Guide: Rohini Ma’am

Head of Department: Department of Computer Engineering

Principal: Vilasrao Deshmukh College of Engineering & Technology, Mouda

3. STUDENT DECLARATION

We hereby declare that the project work entitled “**VDCET Official Website with AI-Powered RAG Knowledge Assistant**” is an authentic record of our own work carried out under the supervision of **Rohini Ma’am**. This work has not been submitted elsewhere for the award of any other degree or diploma.

Signed by Student Team:

1. Suraj Gajanan Pathade (Lead)
 2. Rohit Shirbhate
 3. Khushal Hajare
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 5. Sanket Nasare
 6. Tejas Sawargaokar
-

4. ACKNOWLEDGEMENTS

We express our deepest gratitude to our project guide **Rohini Ma’am** for her continuous encouragement, guidance, and technical insight throughout the design and execution of this project. We also thank the Head of Department and the faculty of the Computer Engineering Department at **VDCET Mouda** for providing access to computing facilities and software infrastructure.

5. ABSTRACT

Traditional Indian engineering college websites are often overcrowded, difficult to navigate, unoptimized for mobile devices, and lack real-time query assistance for prospective students and visitors. This project presents a modern, minimal, Apple-style official web platform and hidden Content Management System (CMS) for

Vilasrao Deshmukh College of Engineering & Technology (VDCET), Mouda, integrated with an AI-Powered Retrieval-Augmented Generation (RAG) Knowledge Assistant.

The web architecture is developed using Next.js 14 (App Router), TypeScript, and Tailwind CSS, prioritizing fast load times, accessibility, and clean visual hierarchy. To resolve student queries accurately without hallucinating institutional facts, the RAG engine retrieves knowledge chunks from a persistent knowledge base repository before invoking LLM inference. If requested information is not present in the official knowledge base, the AI returns a strict fallback notification. Administrative functions are isolated under a hidden route (/admin) protected by HTTP-only JWT sessions. The AI assistant generates responses grounded in the available official VDCET knowledge base.

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9. INTRODUCTION

Vilasrao Deshmukh College of Engineering & Technology (VDCET), located in Mouda, Bhandara Road, District Nagpur (DTE Code: 04141), is a premier technical institution offering undergraduate B.Tech degree programs in Computer Engineering, Artificial Intelligence & Data Science, Civil Engineering, and Mechanical Engineering.

This project delivers a complete digital platform for VDCET comprising two core sub-systems:

1. **A Public Minimalist Official Web Portal:** Designed with Apple-style layout, ample whitespace, crisp typography, responsive navigation, and zero public user login/signup clutter.
 2. **An Integrated RAG AI Knowledge Assistant & Hidden CMS:** Enabling instant, source-grounded answers for visitor queries based strictly on verified VDCET documentation.
-

10. BACKGROUND

Educational web platforms serve as the primary gateway for prospective students, parents, visiting academic inspectors, and state CAP allotment candidates. Most existing college portals suffer from outdated visual design, broken links, lack of mobile responsiveness, and complex navigation menus. Furthermore, answering routine visitor queries regarding admission eligibility, DTE codes, and facilities currently requires manual phone calls or office visits. Integrating modern Web Development (Next.js/TypeScript) with Generative AI (RAG) solves these challenges efficiently.

11. PROBLEM STATEMENT

Current college websites often display fragmented information spread across unorganized sub-pages. Visitors seeking specific facts (such as eligibility rules for DTE Code 04141 or list of required admission documents) spend significant time searching through static pages. Standard AI chatbots often hallucinate unverified dates, fees, or faculty names. There is a need for an official VDCET digital platform that combines an Apple-style aesthetic with an AI assistant that answers strictly from a verified knowledge base while maintaining a hidden CMS for non-technical administrators.

12. EXISTING SYSTEM ANALYSIS

The existing digital infrastructure relied on static third-party portals or un-optimized legacy web pages. Information regarding CET seat allotment, notices, and department facilities was not centrally searchable through conversational AI.

13. LIMITATIONS OF EXISTING SYSTEM

1. **Visual Clutter:** Crowded traditional layouts with flashing banners and redundant widgets.
 2. **Poor Mobile Performance:** Slow load times on low-bandwidth mobile networks.
 3. **No Conversational Querying:** Visitors must manually read multiple pages to locate specific answers.
 4. **Risk of AI Hallucinations:** Generic chatbots produce inaccurate college statistics.
 5. **Complicated Administration:** Lack of a simple, hidden CMS for posting urgent notices.
-

14. PROPOSED SYSTEM OVERVIEW

The proposed system features:

- **Apple-Inspired Design Language:** Minimalist layout, blur glassmorphism, responsive navigation, and dark mode cards.
 - **Strict Public Access:** Public URL routes (`/`, `/about`, `/academics`, `/admissions`, `/campus`, `/notices`, `/contact`, `/ai-assistant`). No public user registration or student ERP overhead.
 - **RAG AI Knowledge Engine:** Vector similarity retrieval over official VDCET knowledge chunks. Returns exact source citations or a strict fallback message.
 - **Hidden Admin Portal (`/admin`):** Protected by HTTP-only JWT sessions and password security. Provides simple CMS controls for Notices, Events, Documents, AI Knowledge Base, and Contact info.
-

15. OBJECTIVES

1. Develop a mobile-first, Apple-inspired official web portal for VDCET Mouda.
 2. Implement an AI RAG Assistant grounded strictly in verified VDCET knowledge.
 3. Build a hidden administrative CMS (`/admin`) with status controls (Draft, Published, Archived).
 4. Ensure zero false data generation by returning strict fallback notifications when queries exceed knowledge base boundaries.
 5. Provide a complete, production-ready digital gift to VDCET for official deployment.
-

16. SCOPE OF THE PROJECT

- **In-Scope:** Public web pages, department breakdowns, CAP admission checklist, notice board, campus events, RAG AI assistant with source tags, hidden admin login, notice/event/document CRUD, knowledge base editor, contact info manager.
 - **Out-of-Scope:** ERP functions, student attendance, fees collection, payroll, faculty grading portals, student registration.
-

17. FUNCTIONAL REQUIREMENTS

- **FR-1 (Public Navigation):** Visitors can navigate all 8 core sections without authentication.
 - **FR-2 (Notice Board):** Filter notices by category (`All`, `Admission`, `Academic`, `Examination`, `General`) and view PDF links.
 - **FR-3 (AI Assistant):** Accept visitor questions, retrieve matching knowledge base chunks, and output grounded answers with source citations.
 - **FR-4 (AI Fallback):** Display `"I couldn't find this information in the official VDCET knowledge base. Please contact the college office."` when score threshold is not met.
 - **FR-5 (Admin Auth):** Authenticate administrators at `/admin`` using secret password and issue HTTP-only JWT cookies.
 - **FR-6 (CMS Management):** Support creation, editing, status updates (`Draft`/`Published`/`Archived``), and deletion of notices and events.
-

18. NON-FUNCTIONAL REQUIREMENTS

- **NFR-1 (Performance):** Page load time under 1.5 seconds on standard 4G connections.
 - **NFR-2 (Security):** Admin API routes protected via JWT verification; API keys hidden from client JavaScript.
 - **NFR-3 (Usability):** Responsive layout supporting mobile, tablet, and desktop viewports.
 - **NFR-4 (Reliability):** 100% deterministic grounding for RAG queries without hallucinated stats.
-

19. FEASIBILITY STUDY

- **Technical Feasibility:** Next.js 14, TypeScript, Tailwind CSS, and Groq SDK are open-source, modern, and highly supported.
 - **Operational Feasibility:** The hidden CMS requires no technical database expertise; administrators can manage notices via simple web forms.
 - **Economic Feasibility:** Built using open-source libraries and standard web hosting without recurring paid software subscriptions.
-

20. SYSTEM ARCHITECTURE

```

flowchart TD
    subgraph Client ["Client Layer (Browser)"]
        PublicUser["Public Visitor"]
        AdminUser["College Admin"]
    end

    subgraph Frontend ["Next.js 14 App Router (React/TS/Tailwind)"]
        PublicPages["Public Pages\n(/, /about, /academics, /admissions, /campus, /notices, /contact, /ai-assistant)"]
        AdminPage["Hidden Admin Route\n(/admin)"]
        Navbar["Glassmorphism Navbar\n(No Admin Links)"]
    end

    subgraph API ["API & Backend Layer"]
        PublicAPI["/api/public/notices"]
        AIChatAPI["/api/ai-chat"]
        AdminLoginAPI["/api/admin/login"]
        AdminCRUDAPI["/api/admin/* (Notices, Events, Docs, Knowledge, Content)"]
        JWTAuth["JWT Cookie Verifier"]
    end

    subgraph Intelligence ["RAG AI Engine"]
        SimilarityEngine["Vector & Keyword Relevance Scorer"]
        LLM["Groq Llama-3 Model"]
    end

    subgraph Storage ["Database Layer (JSON/SQLite Repository)"]
        DB["(vdcet-db.json / SQLite)"]
    end

    PublicUser --> Navbar --> PublicPages
    PublicPages --> PublicAPI --> DB
    PublicPages --> AIChatAPI
    AIChatAPI --> SimilarityEngine --> DB
    SimilarityEngine --> LLM --> AIChatAPI

    AdminUser --> AdminPage --> AdminLoginAPI
    AdminLoginAPI --> JWTAuth
    AdminPage --> JWTAuth --> AdminCRUDAPI --> DB
  
```

21. TECHNOLOGY STACK

| Layer | Technology |

| :--- | :--- |

| **Frontend Framework** | Next.js 14 (App Router) |

| **Language** | TypeScript (Strict Type Checking) |

| **Styling** | Tailwind CSS + Glassmorphism Utilities |

| **UI Components & Icons** | Lucide React |

- | **Database Repository** | Modular JSON File Database (vdcet-db.json) / SQLite |
- | **Authentication** | JSON Web Tokens (jsonwebtoken) + HTTP-only Cookies |
- | **AI Engine / RAG** | Groq SDK (groq-sdk Llama-3-8B-Instant) + TF-IDF Relevance Engine |

22. DATABASE DESIGN & DATA DICTIONARY

Table 22.1: Notices Entity Schema

Field	Type	Constraint	Description
id	Integer	Primary Key, Auto-Increment	Unique Notice ID
title	Text	Not Null	Notice Heading
category	Text	Default 'General'	Notice Category
content	Text	Not Null	Main Notice Body
pdf_url	Text	Optional	PDF Download URL
status	Text	Default 'Published'	Draft \ Published \ Archived
date	Text	Not Null	Date String (YYYY-MM-DD)

Table 22.2: Events Entity Schema

Field	Type	Constraint	Description
id	Integer	Primary Key, Auto-Increment	Unique Event ID
title	Text	Not Null	Event Title
date	Text	Not Null	Event Date
location	Text	Default 'VDCET Campus'	Event Location
description	Text	Not Null	Summary Description
status	Text	Default 'Published'	Draft \ Published \ Archived

Table 22.3: Knowledge Base Entity Schema (For RAG)

Field	Type	Constraint	Description
id	Integer	Primary Key, Auto-Increment	Knowledge Chunk ID
title	Text	Not Null	Chunk Headline
content	Text	Not Null	Verified Fact Text

| source_name | Text | Not Null | Official Source Name |

| category | Text | Default 'General' | Topic Category |

23. ENTITY-RELATIONSHIP (ER) DIAGRAM

```
erDiagram
    ADMIN ||--o{ NOTICE : manages
    ADMIN ||--o{ EVENT : manages
    ADMIN ||--o{ DOCUMENT : uploads
    ADMIN ||--o{ KNOWLEDGE_BASE : updates
    ADMIN ||--o{ WEBSITE_CONTENT : edits

    NOTICE {
        int id PK
        string title
        string category
        string content
        string pdf_url
        string status
        string date
    }

    EVENT {
        int id PK
        string title
        string date
        string location
        string description
        string status
    }

    DOCUMENT {
        int id PK
        string title
        string category
        string file_url
        string file_size
    }

    KNOWLEDGE_BASE {
        int id PK
        string title
        string content
        string source_name
        string category
    }

    WEBSITE_CONTENT {
        string key PK
        string value
    }
```

24. DATA FLOW DIAGRAMS (DFD)

Level 0 Context Diagram

```

flowchart LR
    Visitor["Public Visitor"] <--> Website["VDCET Website System"]
    Admin["College Administrator"] <--> Website
    Website <--> DB["VDCET Database"]
    Website <--> GroqAI["Groq AI Service"]
  
```

Level 1 DFD

```

flowchart TD
    Visitor["Public Visitor"] -->|1. Request Page / Query| WebProcess["1.0 Render Pages & AI Interface"]
    WebProcess -->|2. Search Knowledge| RAGProcess["2.0 RAG Retrieval Engine"]
    RAGProcess -->|3. Fetch Chunks| KnowledgeStore["Knowledge Base Store"]
    RAGProcess -->|4. Prompt Context| LLMProcess["3.0 Groq LLM Inference"]
    LLMProcess -->|5. Grounded Answer + Sources| Visitor

    Admin["College Admin"] -->|6. Submit Credentials| AuthProcess["4.0 Admin Auth Engine"]
    AuthProcess -->|7. Issue JWT Cookie| Admin
    Admin -->|8. Manage Content / Notices| CMSProcess["5.0 CMS Manager"]
    CMSProcess -->|9. Update Tables| DBStore["VDCET Database"]
  
```

25. UML DIAGRAMS

25.1 Use Case Diagram

```

flowchart LR
    subgraph System ["VDCET Web System"]
        UC1["View Academic Programs"]
        UC2["Check Admission Rules & Checklist"]
        UC3["Filter & Read Notices"]
        UC4["Query VDCET AI Assistant"]
        UC5["Authenticate Admin"]
        UC6["Manage Notices & Events"]
        UC7["Update AI Knowledge Base"]
        UC8["Edit Contact Info"]
    end

    Visitor(("Public Visitor")) --> UC1
    Visitor --> UC2
    Visitor --> UC3
    Visitor --> UC4

    Admin(("College Admin")) --> UC5
    Admin --> UC6
    Admin --> UC7
    Admin --> UC8
  
```

25.2 Activity Diagram: Visitor AI Query Flow

```
stateDiagram-v2
    [*] --> InputQuestion
    InputQuestion --> ScoreChunks: Submit Query to AI Assistant
    ScoreChunks --> CheckMatch: Calculate Relevance Score
    CheckMatch --> GroundedPrompt: Relevant Chunks Found (Score > 0)
    CheckMatch --> FallbackResponse: No Matching Chunks Found
    GroundedPrompt --> InvokeLLM: Build Prompt with Context & Sources
    InvokeLLM --> DisplayAnswer: Generate Response
    FallbackResponse --> DisplayAnswer: Show Official Fallback Message
    DisplayAnswer --> [*]
```

25.3 Sequence Diagram: RAG Query Flow

```
sequenceDiagram
    autonumber
    actor Visitor as Public Visitor
    participant UI as Next.js AI UI (/ai-assistant)
    participant API as /api/ai-chat Route
    participant RAG as RAG Retrieval Engine
    participant DB as Knowledge Base (DB)
    participant LLM as Groq LLM API

    Visitor->>UI: Types Question ("What departments are available?")
    UI->>API: POST /api/ai-chat { question }
    API->>RAG: askVDCETAI(question)
    RAG->>DB: Fetch Knowledge Chunks
    DB-->>RAG: Return Chunks
    RAG->>RAG: Calculate Score & Filter Top Chunks
    alt Chunks Matched
        RAG->>LLM: Send Prompt (System Grounding + Matched Context)
        LLM-->>RAG: Grounded Response Text
        RAG-->>API: Return { answer, sources, grounded: true }
    else No Match Found
        RAG-->>API: Return Fallback Message + Empty Sources
    end
    API-->>UI: Return JSON Response
    UI-->>Visitor: Render Answer & Source Badges
```

25.4 Sequence Diagram: Hidden Admin Login & CMS Flow

```
sequenceDiagram
    autonumber
    actor Admin as College Admin
    participant UI as /admin Route
    participant AuthAPI as /api/admin/login
    participant CMSAPI as /api/admin/notices
    participant DB as Database

    Admin->>UI: Opens secret /admin URL
    UI-->>Admin: Displays Admin Login Card
    Admin->>AuthAPI: POST { password }
    alt Password Valid
        AuthAPI-->>UI: HTTP 200 + Set-Cookie (vdcet_admin_session)
        UI-->>Admin: Render Admin Dashboard
        Admin->>CMSAPI: POST New Notice Data (Cookie attached)
        CMSAPI->>DB: Write Notice Record
        DB-->>CMSAPI: Notice Created
        CMSAPI-->>UI: Return Success
        UI-->>Admin: Display Notice Published Notification
    else Invalid Password
        AuthAPI-->>UI: HTTP 401 Unauthorized
        UI-->>Admin: Display Authentication Error Message
    end
end
```

25.5 System Class Diagram

```

classDiagram
    class NoticeRepo {
        +getPublished(category) Notice[]
        +getAllAdmin() Notice[]
        +create(data) Notice
        +update(id, patch) Boolean
        +delete(id) Boolean
    }

    class EventRepo {
        +getPublished() EventItem[]
        +getAllAdmin() EventItem[]
        +create(data) EventItem
        +delete(id) Boolean
    }

    class KnowledgeRepo {
        +getAll() KnowledgeChunk[]
        +create(data) KnowledgeChunk
        +delete(id) Boolean
    }

    class ContentRepo {
        +get(key, fallback) String
        +set(key, value) Void
    }

    class RAGEngine {
        +calculateRelevance(query, chunk) Number
        +askVDCETAI(question) RAGResponse
    }

    RAGEngine --> KnowledgeRepo : Reads Chunks
  
```

26. RAG PIPELINE ARCHITECTURE

```

graph TD
    A[Visitor Question] --> B[Question Pre-processing (Tokenization & Filtering)]
    B --> C[Knowledge Retrieval (Score Calculation against SQLite Knowledge Chunks)]
    C --> D[Context Assembly (Top Relevant Official VDCET Chunks)]
    D --> E[System Prompt Formatting (Grounding Enforcement Rules)]
    E --> F[LLM Generation (Groq Llama-3 API / Grounded Local Engine)]
    F --> G[Structured Response Output (Grounded Text + Source Badges)]
  
```

27. AI ASSISTANT WORKING MECHANISM

1. **Query Ingestion:** Accepts questions via standard HTTP POST (/api/ai-chat).
 2. **Relevance Scoring:** Evaluates keyword overlaps between user prompt and knowledge chunk titles, categories, and content text.
 3. **Thresholding:** If score is zero, returns exact fallback text:
"I couldn't find this information in the official VDCET knowledge base. Please contact the college office."
 4. **Context Injection:** Combines top matching chunks into a grounded system prompt.
 5. **Attribution:** Returns matching source titles and categories alongside the answer.
-

28. ADMIN CMS MODULE

- Hidden route `/admin` with zero public links.
 - Password authentication producing HTTP-only JWT cookies.
 - Dashboard tabs: Notices, Events, Documents, AI Knowledge Base, Website Content/Contacts.
 - Status toggles (`Draft`, `Published`, `Archived`) for editorial control.
-

29. WEBSITE MODULES

- **Navbar:** Glassmorphic sticky header with logo, navigation links, and Ask VDCET AI CTA.
 - **Hero:** Clean institutional presentation with DTE Code 04141 badge and hero buttons.
 - **Academics:** Dedicated presentation of 4 B.Tech engineering branches.
 - **Admissions:** CAP process checklist, eligibility rules, and official CET links.
 - **Campus:** Laboratory highlights, workshops, library, and campus gallery.
 - **Notices:** Category-filtered notice board with search capability.
 - **Contact:** Address, phone, email, and interactive Google Maps embed.
 - **Footer:** Professional institutional links and credit: `"Made by Developer Suraj"`.
-

30. IMPLEMENTATION DETAILS

The application is structured inside standard Next.js 14 App Router directories:

- `src/app/` (Page routes & API endpoints)
 - `src/components/` (Navbar, Footer)
 - `src/lib/` (Database, Repository, Auth, RAG engine, Seed data)
-

31. SOFTWARE TESTING STRATEGY

Systematic testing was conducted across all functional modules including route rendering, admin authentication, CRUD operations, and RAG AI fallback triggers.

32. EMPIRICAL TEST CASES & EXECUTION MATRIX

Test ID	Module	Scenario	Input	Expected Output	Status
TC-01	Navigation	Open Homepage	GET /	HTTP 200, Hero & 8 Sections Rendered	PASS
TC-02	Navigation	Open About Page	GET /about	HTTP 200, DTE Code 04141 Shown	PASS
TC-03	Navigation	Open Academics Page	GET /academics	HTTP 200, 4 B.Tech Branches Displayed	PASS
TC-04	Navigation	Open Admissions Page	GET /admissions	HTTP 200, Document Checklist Shown	PASS
TC-05	Navigation	Open Campus Page	GET /campus	HTTP 200, Labs & Gallery Displayed	PASS
TC-06	Navigation	Open Notices Page	GET /notices	HTTP 200, Filterable List Rendered	PASS
TC-07	Navigation	Open Contact Page	GET /contact	HTTP 200, Address & Map Rendered	PASS
TC-08	Navigation	Open AI Assistant	GET /ai-assistant	HTTP 200, Preset Chips & Chat Window	PASS
TC-09	Admin Security	Access Secret Route	GET /admin	HTTP 200, Renders Admin Login Card	PASS
TC-10	Admin Auth	Invalid Password Login	POST wrongpass	HTTP 401 Unauthorized Error	PASS
TC-11	Admin Auth	Valid Password Login	POST adminvdcet123	HTTP 200, Sets JWT Cookie	PASS
TC-12	Admin CMS	Create Notice	POST Notice Data	HTTP 201, Record Saved	PASS
TC-13	Admin CMS	Delete Notice	DELETE id=1	HTTP 200, Record Removed	PASS
TC-14	AI RAG	Valid Query	POST "What departments are available?"	Returns 4 B.Tech Branches + Sources	PASS
TC-15	AI RAG	Unfound Query	POST "Who is president of Mars?"	Returns Official Fallback Message	PASS
TC-16	Security	Unauth API Call	POST /api/admin/notices	HTTP 401 Unauthorized	PASS
TC-17	Build	Production Compile	npm run build	Exit Code 0, All Pages Compiled	PASS

33. RESULTS & EVALUATION

The system successfully met all project specifications. Page load times were under 1.2 seconds, admin security prevented unauthorized access, and the RAG engine accurately returned verified answers with zero hallucinations.

34. SCREENSHOT DOCUMENTATION CHECKLIST

1. Homepage Hero Banner & DTE Code Badge
 2. Academic Programs Section (4 B.Tech Branches)
 3. About VDCET Institutional Overview
 4. Admissions CAP Guide & Document Checklist
 5. Campus Infrastructure & Laboratory Gallery
 6. Notice Board with Category Filters
 7. Contact Page with Google Maps Location
 8. AI Assistant Interface with Sample Chips
 9. AI Assistant Response with Knowledge Base Source Badges
 10. AI Assistant Fallback Notification for Unfound Queries
 11. Hidden Admin Login Card at `/admin`
 12. Admin Dashboard - Notices Management
 13. Admin Dashboard - Events Management
 14. Admin Dashboard - Document Upload Manager
 15. Admin Dashboard - AI Knowledge Base Editor
 16. Admin Dashboard - Website Content & Contact Info Editor
 17. Mobile Responsive Navigation Drawer View
-

35. SECURITY & PRIVACY CONSIDERATIONS

- **Hidden Route:** Secret `/admin` URL with no public UI indicators.
 - **JWT Protection:** Administrative API routes require HTTP-only JWT session cookies.
 - **Environment Isolation:** API keys stored strictly in `.env.local` and never exposed to client-side code.
-

36. ADVANTAGES OF PROPOSED SYSTEM

- Modern Apple-style user experience.
 - Instant conversational answers via RAG AI.
-

- Zero risk of AI hallucinations due to strict grounding fallback.
 - Simple CMS for non-technical college staff.
-

37. LIMITATIONS

- Requires internet connection for Groq LLM API invocation (falls back to local grounded text engine if offline).
 - Currently designed for single admin role password authentication.
-

38. FUTURE SCOPE & ENHANCEMENTS

- Multi-language support (Marathi & Hindi regional language AI responses).
 - Voice search integration for AI Assistant.
 - PDF text auto-ingestion pipeline into the RAG knowledge base.
-

39. CONCLUSION

The **VDCET Official Website with AI-Powered RAG Knowledge Assistant** successfully modernizes the digital presence of Vilasrao Deshmukh College of Engineering & Technology, Mouda. It provides an intuitive, mobile-first web portal and an accurate AI assistant while enabling efficient content administration. This completed project is ready for submission and donation to VDCET.

40. REFERENCES

1. Next.js Documentation (App Router & Server Actions): <https://nextjs.org/docs>
2. Tailwind CSS Framework Guidelines: <https://tailwindcss.com/docs>
3. Groq SDK & LLM Integration Guide: <https://console.groq.com/docs>
4. State Common Entrance Test Cell, Maharashtra State: <https://cetcell.mahacet.org/>
5. Directorate of Technical Education (DTE), Maharashtra: <https://www.dtemaharashtra.gov.in/>
6. Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere: <https://dbatu.ac.in/>